CSE 2046 Final

22. 06. 2021

2 Sessions (55 minutes each)

An A4 Sized Cheat Sheet is allowed

“On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues”

Take it easy!

Q1. (22 pts) a) How many character comparisons will the Horspool's algorithm make in searching for the pattern 001000 in the binary text of n elements with alternating zeros and ones? 010101….01 (Show shift table). Give your answer in terms of n.

b) Repeat (a) for Boyer Moore Algorithm. Show good suffix and bad symbol tables.

c) Consider the same text of alternating zeros and ones.

(i) Give a pattern of length 4 such that Horspool's algorithm yields less number of comparisons compared to Boyer-Moore.

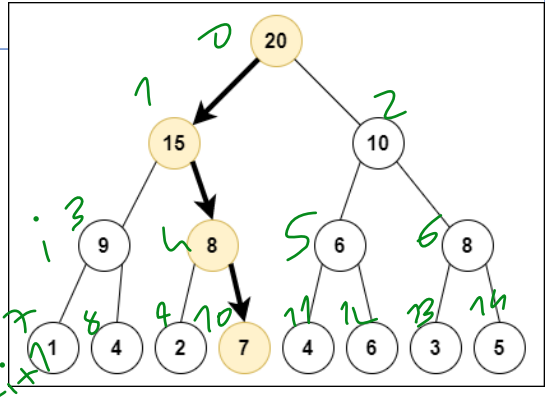
(ii) Give a pattern of length 4 such that Boyer-Moore algorithm yields less number of comparisons compared to Horspool's.

Q2. (23 pts) Consider a max-heap with n nodes. The problem is to find the maximum value path from the root to one of the leaf nodes, i.e. the sum of the values on the visited vertices would be maximized. In the following instance, maximum value path is shown by the arrows.

(a) Describe a dynamic programming algorithm for this problem. Apply your algorithm to the given instance. Give the time and space complexity of your algorithm.

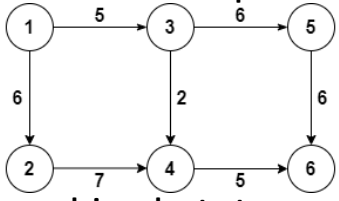
Notes: Heap is represented by an array A. You should give a recurrence relation for MVP(i) which is the maximum value path from i'th entry of the heap array to a leaf node. Describe not only how to find the max value, but also the path.

(b) What is the trivial lower bound of this problem? Is this lower bound tight? Describe.

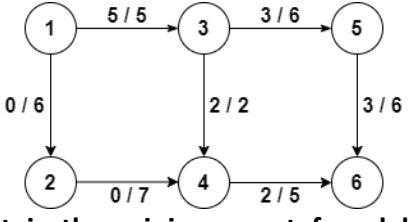


Q3. (~~18~~ 20 pts) Consider the problem described in Q2. Describe a simple greedy algorithm for the same problem and give its time complexity. Give the accuracy ratio of your algorithm for the given instance. Describe an upper bound on the performance ratio of your algorithm for a complete heap with n nodes. Note that the heap may include repeated values.

Q4. (15 pts) Consider the following network. Source is 1 and sink is 6 and the values on links are capacities.



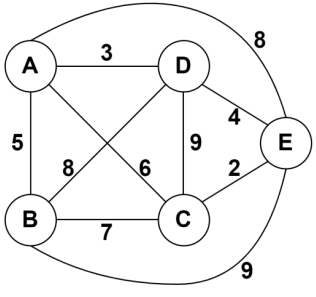
We are applying shortest augmenting path algorithm to find the maximum flow in this network. Following is an intermediate step of the algorithm. Continue from this step and find the maximum flow. Show the labels and BFS queue clearly.



What is the minimum cut found by the algorithm?

Q5. (18 pts) Consider a modified version of Traveling Salesman problem (TSP). Again there are n cities and cost values between cities. The traveling salesman should start from a specified city (say city A) and he needs to visit all cities EXCEPT ONE, and go back to city A. The aim is to find the tour (of length n-1) with minimum total cost.

Define a bounding function and apply branch and bound algorithm for this modified version of TSP on the following graph. SHOW ALL YOUR CALCULATIONS. (Note: You will get minimal grade if you solve ordinary TSP.)



Q6. (14 pts) (a) Given an ASCII coded file consisting of n different characters with various frequencies. The problem is to decide whether it is possible to compress this file by more than 25% by assigning variable length prefix-free binary codewords to each of these characters. Describe whether this problem is in P? In NP? Why?

(b) Suppose that Problem X can be reduced to Shortest Path problem and Problem Y can be reduced to Traveling Salesman problem. All reductions can be done in linear (O(n)) time. For each of the following, indicate whether it is True, False, or Unknown. Give your reasoning, otherwise your answer will not be graded.

(i) Problem Y is more difficult than Prob. X.

(ii) Traveling Salesman problem can be reduced to Problem X in linear time.

(iii) Both Problem X and Problem Y can be tractable problems.

Bonus (10 pts): Prove that decision version of modified TSP in Q5 is NP-Complete.

Q1 a) 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 . . . 0 1

0 0 1 0 0 0

0 0 1 0 0 0

0 0 1 0 0 0

0 0 1 0 0 0

Shift Table

0 → 1

1 → 3

2PTS

1 comparison 3 shift } repeats

2 " 1 shift }

3 comparison 4 shift

3PTS

approximately $\frac{3n}{4}$ comparisons 2PTS

↳ This answer will be accepted.

$\frac{3n-5}{4}$ is also accepted.

b) 0 1 0 1 0 1 0 1 0 1 0 1 0 1 . . . . . 0 1

0 0 1 0 0 0

0 0 1 0 0 0

0 0 1 0 0 0

0 0 1 0 0 0

Bad symbol

0 → 1

1 → 3

Good suffix

k		d_2
1	0 0 1 0 0 0	2
2	0 0 1 0 0 0	1
3	0 0 1 0 0 0	4
4	0 0 1 0 0 0	4
5	0 0 1 0 0 0	4

4PTS

$d_1 = 3-1 = 2$

$d_2 = 2$

so shift by 2.

3PTS

2PTS

n − 6 comparisons will be made.

if n is even, it is n − 7.

c) i. 1000

For Horspool's → 3 comparison 4 shift.

For Boyer Moore → 2 comparison 2 shift.

3PTS

Boyer Moore does more comparison.

See the last page for the other examples.

ii. 1101

Horspool's → 4 comparison 2 shift.

Boyer Moore → 4 comparison 3 shift

1 comparison 1 shift.

5 comp. 4 shift.

3PTS

Q2 $MVP(1) = A[1] + \max\{MVP(2), MVP(3)\}$

in general.

$$MVP(i) = A[i] + \max\{MVP(2i), MVP(2i+1)\} \quad \text{if } i \le \lfloor \frac{n}{2} \rfloor$$

(internal nodes) (6 PTS)

Leaf nodes: (if $i > \lfloor \frac{n}{2} \rfloor$)

$MVP(i) = A(i)$ (4 PTS)

DP algorithm starts from last parental node & calculate and store MVP(i) by decreasing i one by one. MVP(1) is the result.

if we apply this algorithm to the given instance:

$MVP(8) = 1$	$MVP(12) = 4$
$MVP(9) = 4$	$MVP(13) = 6$
$MVP(10) = 2$	$MVP(14) = 3$
$MVP(11) = 7$	$MVP(15) = 5$

(5 PTS)

$MVP(7) = 8 + \max\{3, 5\} = 13$

$MVP(6) = 6 + \max(4, 6) = 12$

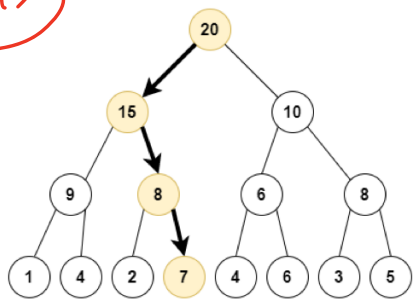
$MVP(5) = 8 + \max(2, 7) = 15$

$MVP(4) = 3 + \max\{1, 4\} = 7$

$MVP(3) = 10 + \max\{7, 12\} = 22$

$MVP(2) = 15 + \max\{7, 15\} = 30$

$MVP(1) = 20 + \max\{30, 22\} = 50.$



Then we can go backwards to find the path. (2 PTS)

Time complexity : $\Theta(n)$. Space complexity : $\Theta(n)$. (2 PTS)

b) Trivial LB is $\Omega(n)$ because we may need to process all the values. It is tight, because the above algorithm is also $\Theta(n)$.

(4 PTS)

Q3 Simple greedy alg:

Start from the root node $i \leftarrow 1$;

while $2i \leq n$ or $2i+1 \leq n$ // children are in the heap.

if $A[2i] \geq A[2i+1]$

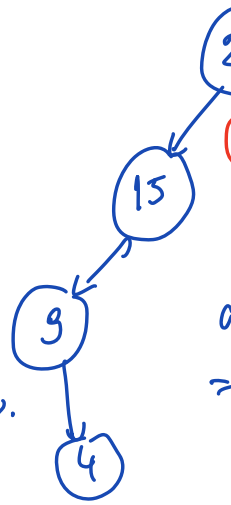
$i \leftarrow 2i$

else

$i \leftarrow 2i+1$

mark(i).

4 PTS



3 PTS

accuracy ratio $= \frac{50}{48}$

3 PTS

Time complexity:

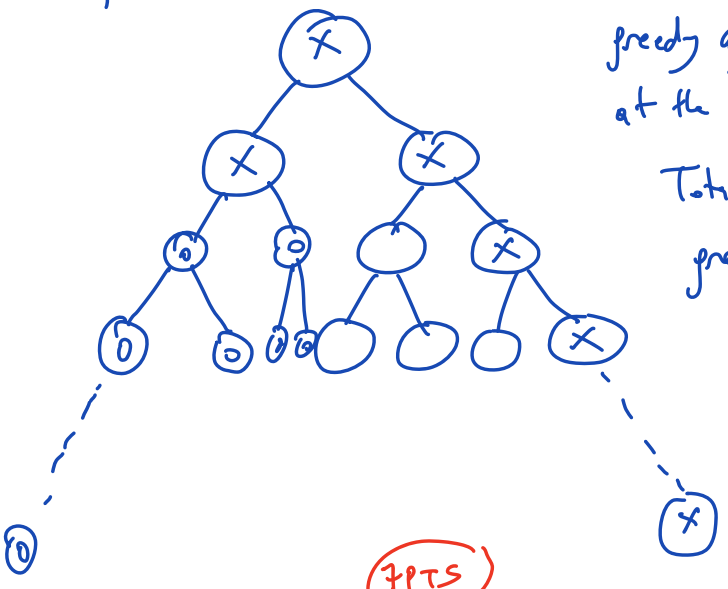
We need $3 \times$ (heigh of the heap) comparisons.

Height of the heap is $\log_2 n$.

Therefore time complexity is $\Theta(\log n)$.

3 PTS

Worst performance ratio is obtained in such an instance:



7 PTS

greedy algorithm chooses left child at the first step.

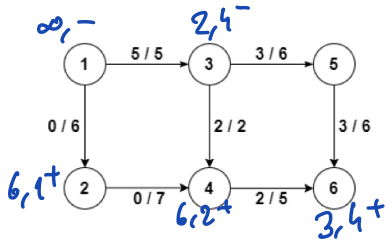
Total value obtained by the greedy alg $= 2X$

optimal result $= \log_2 n \cdot X$

upper bound on the performance ratio is

$$\frac{\log_2 n \cdot X}{2 \cdot X} = \log_2 \sqrt{n}.$$

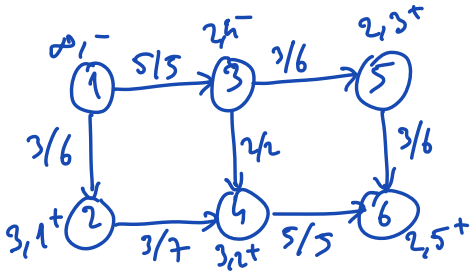
Q4



Queue: 1 2 4 3 6

aug. flow: 1 → 2 → 4 → 6

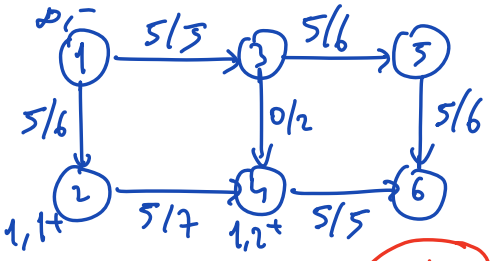
flow: 3 (5 pts)



Queue: 1 2 4 3 5 6

aug. flow: 1 → 2 → 4 ← 3 → 5 → 6

flow: 2 (5 pts)

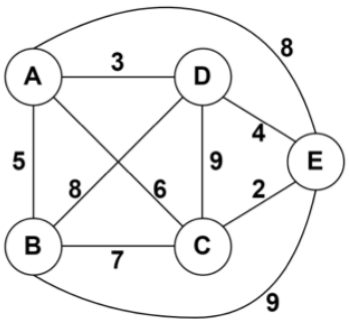


Queue: 1 2 4 (2 pts)

Total flow: 10 //

min cut: {(1,3), (4,6)} (3 pts)

Q5



start

u/o D: lb: $\lceil \frac{3+6+6+2+3+4+2+4}{2} \rceil$ = 15

u/o C: lb: $\lceil \frac{3+5+5+8+3+4+4+8}{2} \rceil$ = 20 X

u/o D: lb: $\lceil \frac{5+6+5+7+6+2+2+8}{2} \rceil$ = 21 X

u/o E: lb: $\lceil \frac{3+5+5+7+6+7+3+8}{2} \rceil$ = 22 X

AC: lb: $\lceil \frac{3+6+6+2+3+4+2+4}{2} \rceil$ = 15

AD: lb: $\lceil \frac{3+6+6+2+3+4+4+2}{2} \rceil$ = 15 X

AE: lb: $\lceil \frac{8+3+6+2+3+4+8+2}{2} \rceil$ = 18 X

ACDEA
cost: 27 first sol

ACEDA
cost: 15 optimal sol

Q6 a) This problem can be solved by Huffman coding algorithm.
Huffman coding is a polynomial time algorithm.
Therefore the problem is in P. (3PTS)
P ⊆ NP, therefore it is in NP as well. (2PTS)

b) Problem X ⟶ Shortest Path
Problem Y ⟶ TSP

i) With the above knowledge, we know that: (3PTS)
- Shortest Path is at least as difficult as Problem X.
- TSP " " " " " Problem Y.
But we can't compare Problem X & Y. UNKNOWN

ii) Problem X has an algorithm with $O(n^2)$ time complexity (Dijkstra)
If TSP can be reduced to Problem X in linear time, this means that TSP has also $O(n^2)$ algorithm. However TSP is an NP-Complete problem. FALSE (3PTS)
Note: There may be a polynomial time alg for TSP, if P=NP. But possibly it will not be $O(n^2)$. Anyway, I accepted answers which says UNKNOWN, because we don't know whether P=NP.

iii) Of course both can be tractable.
It is clear that Problem X is tractable. (3PTS)
Problem Y can be reduced to TSP, but may be there can be other algorithms with polynomial time complexity. TRUE
Answers saying Problem X tractable, Prob Y unknown are OK.

1c) 0|0|0|0|0|0|

0000 HP: 1C4S BM: 1C4S same.

0001 HP: 3C4S Bm: 3C4S same.

0010 HP: 1C1S + 4C2S BM: 1C1S + 4C3S ⇒ 5C4S BM better.

0011 HP: 2C1S + 1C2S BM: 2C1S + 1C2S same.

0100 HP: 1C2S BM: 1C2S same.

0101 same.

0110 HP: 1C1S + 3C3S ⇒ 4C4S BM: 1C1S + 3C3S ⇒ 4C4S same.

0111 HP: 2C1S + 1C3S ⇒ 3C4S BM: 2C2S HP better.

1000 HP: 1C3S + 2C1S ⇒ 3C4S BM: 1C3S + 2C2S HP better.

1001 HP: 3C3S + 1C1S ⇒ 4C4S BM: 3C3S + 1C1S same.

1010 HP: 1C1S BM: 1C1S same.

1011 HP: 2C1S + 1C2S BM: 2C1S + 1C2S same.

1100 HP: 1C2S BM: 1C2S same.

1101 HP: 4C2S BM: 4C3S + 1C1S ⇒ 5C4S BM better

1110 HP: 1C1S + 3C4S BM: 1C1S + 3C4S same.

1111 HP: 2C1S + 1C4S BM: 2C3S + 1C4S BM better.